



**HỌC VIỆN  
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# **Web Design**

## **Lecture 1 - Introduction to Web design**



Name all the terminologies you know about Web Development

E.g. Website, web page, HTML, Google Chrome,



- 1.1. Web design basic concepts
- 1.2. Web development process
- 1.3. Related tasks
- 1.4. Web design languages
- 1.5. Web design tools

## 1.1. Web design basic concepts



- The Internet vs. the web
- Serving up your information
- Browsers
- Web page addresses (URLs)
- The anatomy of a web page
- Putting it all together



The internet is an **international network of connected computers**.

No company owns the internet; it is a cooperative effort governed by a system of standards and rules.

There are many ways information can be passed between computers, including *email (POP3/IMAP/SMTP), file transfer (FTP), secure shell (SSH),... (protocols)*.

The web (originally called the **World Wide Web**, thus the “www” in site addresses) is just one of the ways information can be shared over the internet.

It is unique in that it allows *documents to be linked to one another via hypertext links*—thus forming a huge “web” of connected information.

The web uses a protocol called **HTTP (HyperText Transfer Protocol)**.

The web is a **subset of the internet**. It is just one of many ways information can be transferred over networked computers.



The **server** is the software (not the computer itself) that *allows the computer to communicate with other computers*; (however, it is common to use the word “server” to refer to the computer as well).

The role of server software is to ***wait for a request for information, and then retrieve and send that information back*** as quickly as possible.

**Terminologies:** HTTP, Apache, IIS, IP Address, DNS → Which one do you know?



The software that does the requesting is called the ***client***.

People use desktop browsers, mobile browsers, and other assistive technologies (such as screen readers) as clients to access documents on the web. The server returns the documents for the browser (also referred to as the user agent in technical circles) to display



What browsers do you know? Which one do you prefer? Why?

Try accessing some websites on a browser from both the computer and from the mobile device, any differences?

E.g. <https://dantri.com.vn/> <https://www.youtube.com/> ...

*Tip: turn on the responsive view in the inspect tool to see all the screen size*



Indicate which machine is doing the processing.

- Client-side applications run on the user's machine (also referred to as the **frontend**)
- While server-side applications and functions use the processing power of the server computer (the **backend**).



A brief history of the Internet

Intranets and Extranets

Open source

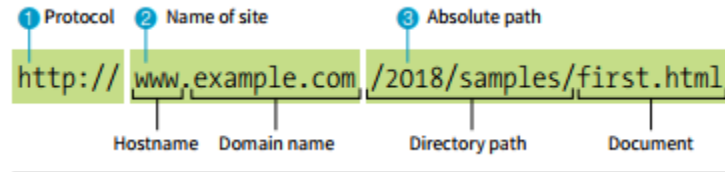
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Every page and resource on the web has its own special address called a URL, which stands for Uniform Resource Locator.

### ***The Parts of a URL***

A complete URL is generally made up of three components: ***the protocol, the site name, and the absolute path*** to the document or resource





Open <https://dantri.com.vn/> and browse some pages inside it to see the elements of the URL

Open <https://vnexpress.net/> and browse some pages, similarly



## Simplified URLs

- Skipping the protocol (http and https)
- Pointing to default files: When a server receives a request for a directory name rather than a specific file, it looks in that directory for a default document, typically named ***index.html***

## Shortened URLs



HTML Documents

Pictures

Styles

Behavior





Safari: Develop → Show → Page Source

Chrome: View → Developer → View Source

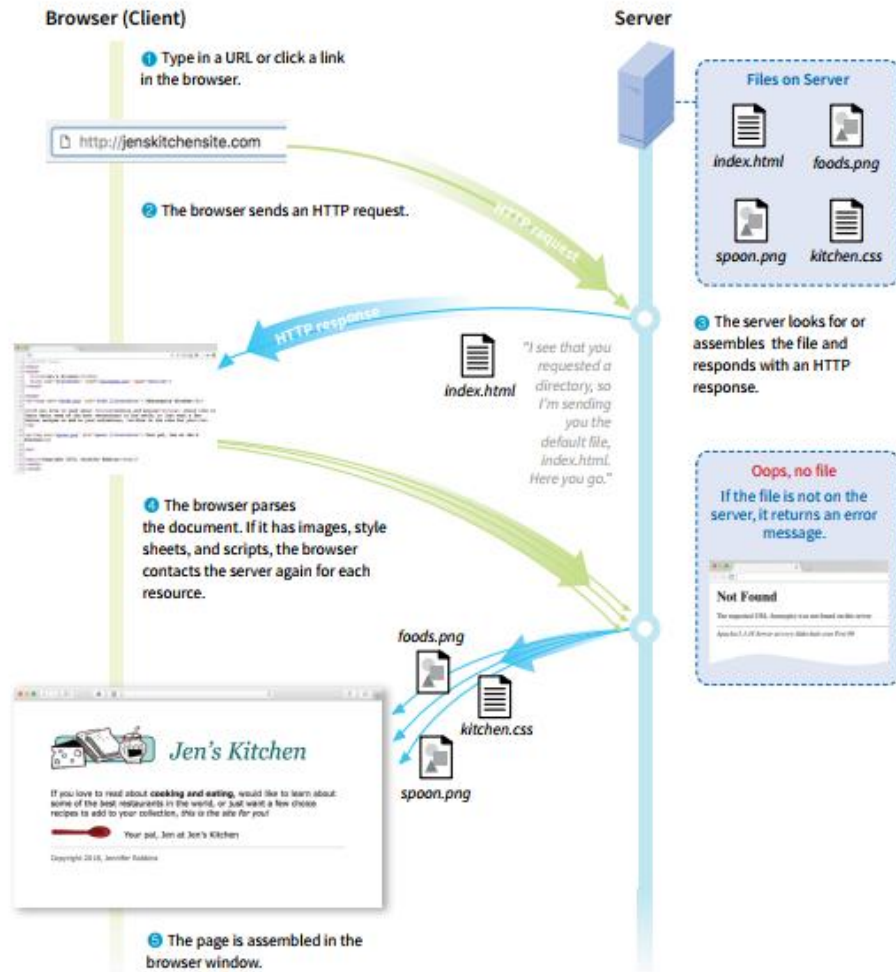
Firefox: Tools → Web Developer → Page Source

Or just press the F12 button

Inspect the image, menu, title, ... you see in the page

## Putting it all together

<https://www.youtube.com/watch?v=0cU4sIP-OuE>





Static vs. Dynamic Sites

HTTP Status Codes

Self reading, page 32



1. HTML \_\_\_\_\_ a. The software that wait for request and return documents
2. W3C \_\_\_\_\_ b. The location of a web document or resource
3. Server \_\_\_\_\_ c. The markup language used to describe web content
4. CSS \_\_\_\_\_ d. Matches domain names with numeric IP addresses
5. HTTP \_\_\_\_\_ e. A protocol for file transfer
6. IP \_\_\_\_\_ f. Protocol for transferring web documents on the internet
7. URL \_\_\_\_\_ g. The language used to instruct how web content looks
8. IIS \_\_\_\_\_ h. A web-server software provided by Microsoft
9. DNS \_\_\_\_\_ i. Internet Protocol
10. FTP \_\_\_\_\_ j. The organization that monitors web technologies

## 1.2. Web development process



1. Requirements collection
2. Planning
3. Designing
4. Content creation
5. Programming
6. Testing
7. Implementation & maintenance



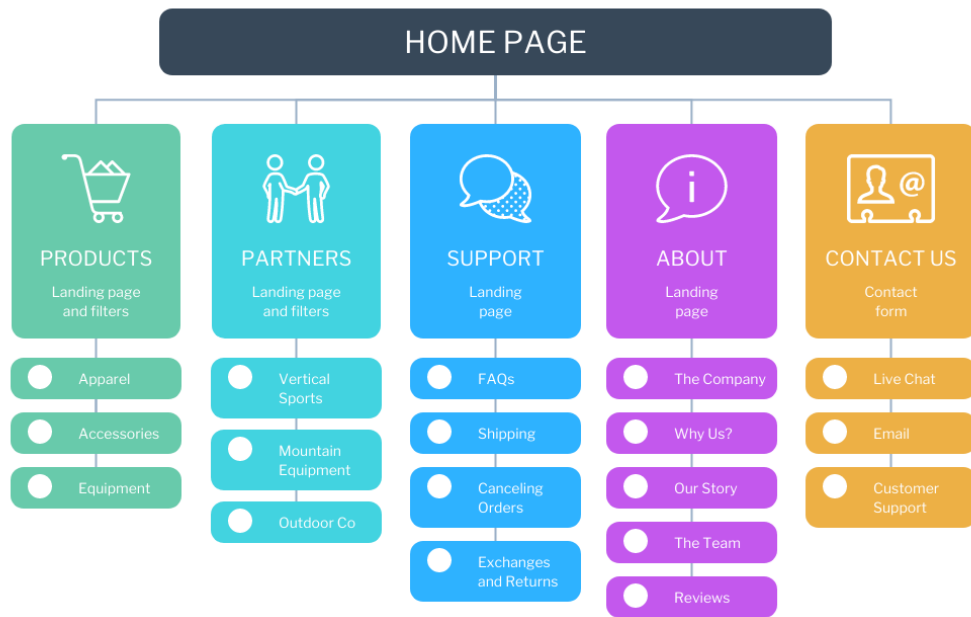
Interview the customer, analyze the documents to understand the requirement:

- Purpose of the website
- Target audience
- Main features of the website
- Technology, theme
- Budget



Make a detail plan including the purpose, budget, timeline, resources

Build a **sitemap / wireframe**



Footer (primary navigation + additional)

- Store Locations
- News
- Advice

- Sector 1
- Sector 2
- Sector 3

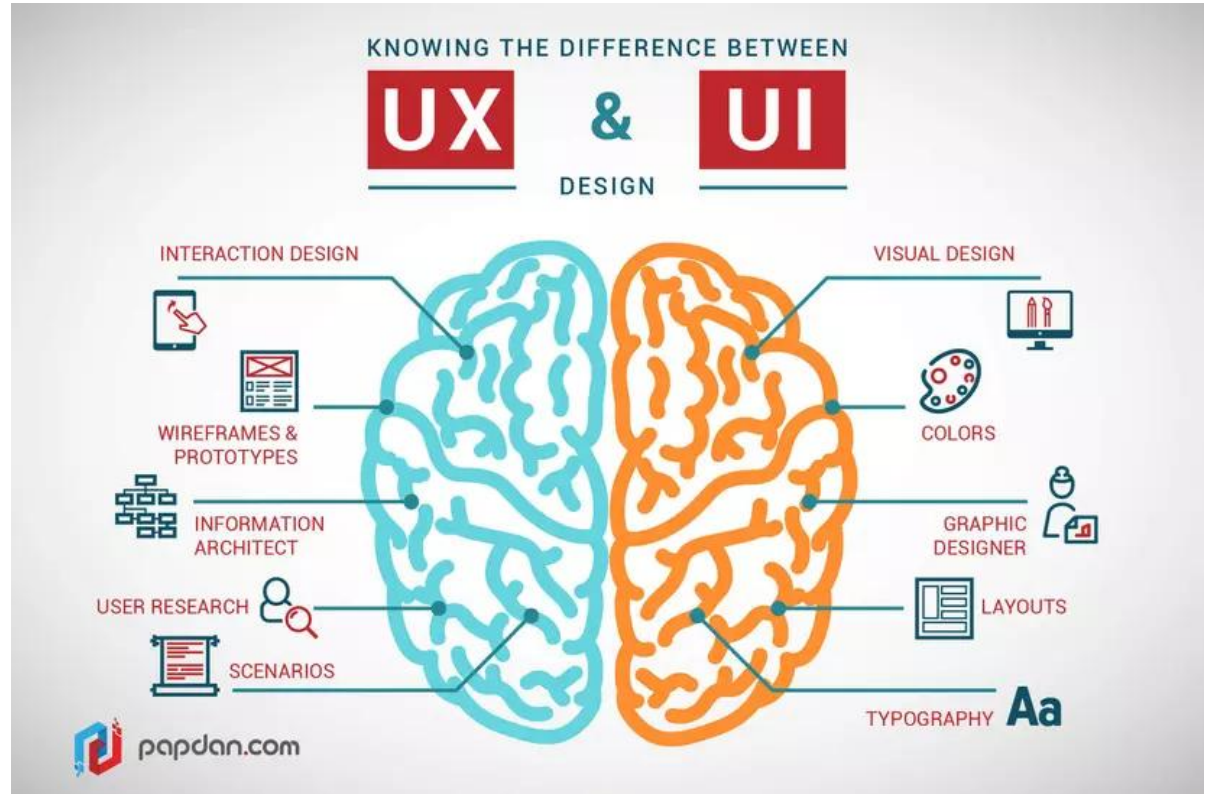
- Product
- Weekly Deals
- Seasonal Clearance

- Newsletter Sign Up
- Terms of Business

- Facebook
- Instagram
- Twitter



Including both  
UI & UX:





### ***Content is king***

Content could be the About Us page, the product description page, the menu title

Normally, customers will provide the content



Write source codes to make the website works





Verify that all the visual and functional parts of the website work normally

- Functional test
- Availability test
- GUI test
- Compatibility test
- Performance test
- Security test



Putting the source code & necessary files on a web server to make the website runs on the Internet

Tracking the performance of the website

Error handling



Content creator

UI/UX designer

Visual designer

**Frontend developer**

Backend developer

SEO

Product manager

Multimedia producers



What are the differences among these job titles

UI/UX designer

Visual designer

**Frontend developer**

Discussion



HTML, CSS, JAVASCRIPT

How about: PHP, JSP, ASP.Net?



HTML stands for Hyper Text Markup Language

HTML is the standard markup language for creating Web pages

HTML describes the structure of a Web page



CSS is the language we use to style a Web page.

CSS stands for Cascading Style Sheets

CSS describes how HTML elements are to be displayed on screen, paper, or in other media



JavaScript is the world's most popular programming language.

JavaScript is the programming language of the Web.



For user interface designing:

- Adobe
  - Adobe Dreamweaver
  - Adobe Photoshop
  - Adobe Illustrator
- Figma



For coding:

- **VS Code**
- Sublime text
- JetBrains
- Codepen.io
- jsFiddle.net



For debugging:

- View page source (Ctrl + U)
- Developer tools on browsers (F12)
  - Responsive view



Website builder (no-code):

- Google site
- MS Power pages
- <https://bubble.io/>
- <https://www.shopify.com/website/builder>
- <https://www.wix.com/>

***Find a free website builder platform and try building your favourite website***



Install your favourite coding tools

Find and participate in Front-end & web development communities (Facebook, Reddit, Discord, Slack, ...)

Name at least 10 different web design roles